

HELSINKI HARMAJA, Finland

WMO: 027950

Lat: 60.105N

Lon: 24.976E

Elev: 21

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.6	8.0	-1.5	5.1	2.7	4.2	6.8	8.9	36.9	32.0	35.0	32.5	11.5	20	0.774

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	74.3	68.7	71.8	66.7	69.2	64.4	69.6	73.4	67.8	71.5	65.0	68.3	11.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	103.5	72.4	66.2	96.5	71.1	63.6	87.9	67.6	33.6	73.7	32.2	71.8	30.0	68.2	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
33.4	30.1	27.3	0.8	75.5	7.4	4.8	-4.5	78.9	-8.8	81.7	-13.0	84.4	-18.3	87.8		
			1.7	69.7	8.4	3.9	-4.4	72.5	-9.3	74.8	-14.0	77.0	-20.2	79.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.7	25.3	26.6	30.9	38.0	48.1	55.5	63.1	62.1	55.9	45.3	39.0	32.9
	DBStd	14.19	8.65	9.00	5.27	3.76	5.46	4.02	5.30	4.08	4.68	5.12	6.03	7.36
	HDD50	3510	766	655	592	361	103	4	0	0	11	156	330	532
	HDD65	7856	1231	1075	1057	811	524	285	105	109	272	610	780	997
	CDD50	1195	0	0	0	0	44	170	405	376	189	11	0	0
	CDD65	65	0	0	0	0	0	0	45	20	0	0	0	0
	CDH74	77	0	0	0	0	0	1	63	13	0	0	0	0
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0
	WSAvg	13.9	15.1	14.4	14.2	11.7	11.6	12.7	11.2	12.0	13.8	15.7	16.5	18.4
	PrecAvg	23.7	1.7	1.3	1.2	1.3	1.4	2.0	2.2	2.9	2.4	2.7	2.6	2.0
Precipitation	PrecMax	33.0	3.3	3.9	2.2	2.8	2.9	4.9	6.0	6.5	6.1	7.0	6.7	5.0
	PrecMin	14.2	0.0	0.0	0.2	0.3	0.2	0.4	0.1	0.0	0.7	0.6	0.2	0.2
	PrecStd	4.0	0.9	0.9	0.6	0.7	0.8	1.1	1.6	1.6	1.3	1.4	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	39.4	41.8	51.6	65.3	67.5	78.0	75.6	65.6	56.3	49.5	45.7
		MCWB	39.3	35.7	37.0	45.2	56.5	61.5	70.5	69.6	62.0	52.8	47.6	44.4
	2%	DB	39.2	37.1	39.5	47.4	61.7	64.9	75.7	72.5	63.9	54.3	47.9	44.0
		MCWB	38.2	35.8	36.6	42.7	55.1	60.1	69.6	67.4	60.8	51.8	46.8	42.6
	5%	DB	37.6	36.1	37.8	45.1	58.8	63.3	73.6	70.1	62.9	52.9	46.7	42.5
		MCWB	36.7	35.1	35.7	41.1	53.6	58.9	68.2	65.1	59.8	50.8	45.4	41.0
	10%	DB	35.7	35.2	36.6	43.1	55.7	61.6	71.4	68.0	61.8	51.8	45.7	40.9
		MCWB	34.8	34.4	34.9	40.0	51.8	57.6	66.3	63.3	58.8	49.8	44.4	39.5
	0.4%	WB	39.7	37.3	38.5	45.6	58.3	63.9	72.7	70.7	63.4	54.2	48.4	45.1
		MCDB	40.0	37.7	40.1	50.1	63.0	66.3	75.3	74.7	64.8	55.6	48.9	45.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.4	36.2	37.3	43.6	56.1	61.2	70.8	68.5	61.9	52.9	47.1	43.0
		MCDB	38.9	36.7	38.6	47.1	60.7	64.1	74.3	71.8	63.5	53.9	47.8	43.9
	5%	WB	36.9	35.4	36.3	41.9	54.1	59.7	69.1	66.5	60.7	51.7	45.9	41.2
		MCDB	37.6	36.0	37.5	44.5	58.1	62.7	73.1	69.6	62.4	52.6	46.6	42.2
	10%	WB	35.1	34.6	35.3	40.5	52.2	58.2	67.4	64.3	59.4	50.3	44.7	39.7
		MCDB	35.8	35.2	36.4	42.6	55.7	60.9	71.3	67.0	61.2	51.6	45.7	40.8
Mean Daily Temperature Range	5% DB	MDBR	6.4	5.5	5.4	5.5	6.5	6.5	6.5	6.0	5.5	5.1	4.6	5.4
		MCDBR	4.8	3.9	5.5	8.5	9.6	7.9	7.9	6.7	6.2	5.0	4.4	5.5
		MCWBR	4.7	3.5	4.1	5.0	5.8	5.0	4.9	4.7	4.4	4.6	4.6	5.7
	5% WB	MCDBR	4.5	3.8	5.0	7.7	9.0	7.0	6.9	5.5	5.3	4.6	4.4	4.7
		MCWBR	4.5	3.5	3.9	4.9	6.1	5.2	4.7	4.6	4.3	4.9	4.7	5.1
Clear-Sky Solar Irradiance	taub		0.280	0.313	0.335	0.360	0.359	0.357	0.379	0.373	0.345	0.334	0.308	0.265
	taud		2.146	2.194	2.248	2.334	2.411	2.480	2.441	2.456	2.520	2.396	2.218	2.094
	Ebn at Noon		151	213	250	266	276	278	268	260	249	208	139	107
	Edh at Noon		14	23	30	33	33	31	32	29	23	19	13	10
All-Sky Solar Radiation	RadAvg		79	259	732	1222	1673	1739	1676	1301	792	345	96	47
	RadStd		14	49	94	128	180	135	151	133	103	53	17	7

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (59 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	-247	N/A	N/A	N/A	

Nomenclature: See separate page